Jose Antonio Charles

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Chemical Engineer with process safety and risk assessment experience in refining and nuclear industries, applying PHA, HAZOP, and MOC under PSM and CalARP to identify initiating events, evaluate safety controls, and develop risk measures.

Education

California State University, Long Beach
Master of Science in Chemical Engineering — May 2025
Bachelor of Science in Chemical Engineering | Minor in Chemistry — May 2022

Skills

Software: MS Office Suite (Outlook, Word, Excel, Teams), SolidWorks, Bluebeam Revu
Process Simulation: Aspen Tech (HYSYS/Plus), AVEVA PRO/II, MATLAB
Technical: Geometric Dimensioning and Tolerancing (GD&T)
Professional: Project Management, Technical Liaison, Integrated Safety Analysis (ISA), Safety Basis Documentation
Process Safety: Process Hazard Analysis (PHA), HAZOP, What If/Checklist, Management of Change (MOC), CalARP, Risk Management Plan (RMP), Offsite Consequence Analysis (OCA)

Professional Experience

Hukari Ascendent | El Segundo, CA **(Nuclear Facility Safety)** — February 2026 – August 2026
Nuclear Safety Engineer (Integrated Safety Analysis)

- Lead Integrated Safety Analyses (ISA), PHAs, HAZOP, and What If/Checklist studies across 7 flow control systems, identifying initiating events, controls, and preventive/mitigative functions under DOE ISM and PSM-equivalent requirements.
- Create and maintain an Excel dashboard tracking and risk-ranking action items from PHA and HAZOP recommendations, giving clear visibility into open findings, owners, and status across 7 systems.
- Evaluate performance, reliability, and off-normal/accident conditions across 7 systems, deriving preventive and mitigative controls and producing safety-basis documentation per API, ASME, NFPA, and ANS.

Los Angeles Unified School District (LAUSD) | Los Angeles, CA — October 2025 – February 2026
Associate Project Engineer

- Conducted technical reviews of the engineering and safety aspects of specifications, RFIs, and datasheets on as-built packages to verify accuracy, constructability, and compliance with OSHA and federal, state, and local regulatory requirements.
- Analyzed schedules, change orders, and cost to flag safety and schedule risks, negotiate fixes, and maintain budget and reliability targets.
- Served as technical advisor and liaison among project managers, contractors, and inspection teams, coordinating field inspections, resolving RFIs, clarifying design intent, and documenting compliance.

Marathon Petroleum Corporation | Carson, California — July 2024 – October 2024
HAZID/Management of Change (MOC) Coordinator

- Executed 10+ Management of Change (MOC) initiatives for flow control upgrades and electrification under PSM (29 CFR 1910.119), integrating CMMS updates to streamline preventive maintenance and mechanical integrity.
- Performed application review and risk assessments of customer operating conditions for piping repurposing, overseeing field execution of condensate flow control installation per Gas Standards and process-safety criteria.
- Conducted hazard identification (HAZID) of refinery operations involving acutely hazardous materials (AHM), developing mitigative controls aligned with PSM and California Accidental Release Prevention (CalARP) requirements.

Senior Aerospace SSP | Burbank, California — October 2022 – April 2024
Manufacturing Engineer | *Operations Development Manager*

- Applied Lean Six Sigma and structured analytical, risk-based methods across 11 engineered-to-order manufacturing workflows, performing sizing, selection, and costing activities to reduce variance by 13% and optimize throughput.
- Validated process specifications for fixture redesign, performing root-cause analysis to resolve blueprint tolerance discrepancies and reduce waste by 70%.
- Partnered with manufacturing, quality, and cross-functional teams to streamline ERP/MRP enhancements, BOM tracking, order routings, and compliance documentation, achieving 25% lead-time reduction and improved order efficiency.

Engineering Project

Integrated Refinery Simulation and Optimization | California State University, Long Beach — May 2025
Graduate Process Engineer | (MATLAB, Aspen HYSYS, Excel)

- Simulated a 600,000 kg/h crude fluid handling and preheat train (desalter, exchangers, flash separator), optimizing flow control and unit mass balance across heat recovery systems to improve energy efficiency and reduce heater duty.
- Performed application review and sizing/selection of a centrifugal compressor for a 110 MMSCFD system; designed integrated gas flowsheet (inlet separation, chiller, LTS) with heat exchange optimization, cutting chiller duty by 2.4 MW.

- Designed a refrigerated gas plant flowsheet (inlet separation, chiller, LTS, sales-gas heater) with integrated heat exchange, cutting chiller duty by 2.4 MW and delivering sales gas at 10 degrees Celsius and 6,200 kPa.